

WALPEUP, Australia

WMO: 958310

Lat: 35.1200S

Lon: 142.0039E

Elev: 105

StdP: 100.07

Time Zone: 10.00 (AUS)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|------------|------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |            |            | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>7      | (b)<br>2.4 | (c)<br>3.4 | (d)<br>-6.8                   | (e)<br>2.1 | (f)<br>27.1 | (g)<br>-4.9 | (h)<br>2.5 | (i)<br>25.8 | (j)<br>12.9           | (k)<br>15.0 | (l)<br>11.1 | (m)<br>13.5 | (n)<br>2.5            | (o)<br>200 | (p)<br>0.531 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 15.7                   | 40.1            | 18.9 | 37.6 | 18.2 | 35.3 | 17.7 | 21.4                | 30.7 | 20.4 | 30.3 | 19.6 | 30.0 | 5.4                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.3                            | 14.2 | 23.1 | 17.9 | 13.0 | 22.5 | 16.3 | 11.7 | 22.5 | 62.6          | 30.0 | 59.1 | 30.1 | 56.1 | 29.5 | 24.7              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 11.3              | 9.7  | 8.5 | 0.3                        | 43.9 | 0.9                | 1.5 | -0.3                                               | 45.0 | -0.8       | 45.9 | -1.3       | 46.7 | -1.9       | 47.9 |  |  |
|                   |      |     | -0.1                       | 23.0 | 0.8                | 1.2 | -0.7                                               | 23.8 | -1.2       | 24.5 | -1.6       | 25.2 | -2.3       | 26.1 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 17.1   | 24.7  | 23.9  | 21.2  | 17.4  | 13.5  | 10.6  | 10.1  | 11.0  | 13.5  | 16.7  | 20.1  | 22.4  |
|                                                                        | DBStd       | 6.39   | 4.84  | 4.53  | 4.48  | 3.75  | 2.86  | 2.04  | 2.10  | 2.64  | 3.36  | 4.33  | 4.79  | 4.89  |
|                                                                        | HDD10.0     | 66     | 0     | 0     | 0     | 0     | 3     | 16    | 25    | 18    | 3     | 1     | 0     | 0     |
|                                                                        | HDD18.3     | 1240   | 3     | 4     | 18    | 61    | 153   | 231   | 256   | 228   | 152   | 87    | 34    | 12    |
|                                                                        | CDD10.0     | 2639   | 454   | 389   | 348   | 223   | 110   | 35    | 28    | 48    | 108   | 207   | 303   | 385   |
|                                                                        | CDD18.3     | 772    | 199   | 160   | 108   | 34    | 2     | 0     | 0     | 1     | 7     | 35    | 87    | 139   |
|                                                                        | CDH23.3     | 10378  | 2828  | 2079  | 1272  | 339   | 16    | 0     | 0     | 7     | 109   | 526   | 1216  | 1988  |
| (14) Wind                                                              | WSAvg       | 4.3    | 4.7   | 4.5   | 4.2   | 3.9   | 4.0   | 3.8   | 4.1   | 4.1   | 4.4   | 4.5   | 4.6   | 4.7   |
|                                                                        | PrecAvg     | 290    | 23    | 21    | 11    | 20    | 23    | 28    | 28    | 26    | 31    | 25    | 26    | 28    |
|                                                                        | PrecMax     | 496    | 111   | 106   | 54    | 72    | 80    | 70    | 63    | 55    | 122   | 69    | 70    | 112   |
|                                                                        | PrecMin     | 151    | 0     | 0     | 0     | 0     | 0     | 5     | 6     | 1     | 4     | 0     | 2     | 2     |
|                                                                        | PrecStd     | 93     | 26    | 24    | 13    | 21    | 19    | 16    | 14    | 15    | 26    | 22    | 19    | 27    |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 43.4  | 41.8  | 38.2  | 33.4  | 25.4  | 19.5  | 19.5  | 24.2  | 30.1  | 35.3  | 38.9  | 41.9  |
|                                                                        |             | MCWB   | 19.7  | 20.3  | 18.3  | 17.1  | 14.3  | 12.5  | 11.7  | 12.6  | 14.3  | 15.7  | 17.8  | 18.7  |
|                                                                        | 2%          | DB     | 40.7  | 38.3  | 35.1  | 29.6  | 22.6  | 17.3  | 17.1  | 20.8  | 26.6  | 31.8  | 35.5  | 38.6  |
|                                                                        |             | MCWB   | 19.4  | 18.9  | 17.9  | 15.7  | 13.5  | 11.7  | 11.0  | 11.5  | 13.5  | 15.3  | 17.0  | 17.8  |
|                                                                        | 5%          | DB     | 37.8  | 35.6  | 32.6  | 27.2  | 20.4  | 15.9  | 15.6  | 18.4  | 23.5  | 29.0  | 32.9  | 35.6  |
|                                                                        |             | MCWB   | 18.5  | 18.5  | 17.6  | 15.0  | 13.0  | 11.2  | 10.7  | 10.7  | 12.6  | 14.1  | 16.4  | 17.1  |
|                                                                        | 10%         | DB     | 34.8  | 33.2  | 29.9  | 24.7  | 18.6  | 14.7  | 14.3  | 16.5  | 20.6  | 26.1  | 30.3  | 32.7  |
|                                                                        |             | MCWB   | 17.9  | 18.0  | 17.0  | 14.2  | 12.5  | 10.9  | 10.2  | 10.2  | 11.6  | 13.7  | 15.6  | 16.4  |
|                                                                        | 0.4%        | WB     | 22.7  | 23.2  | 20.7  | 18.6  | 16.2  | 14.3  | 13.6  | 13.7  | 15.5  | 18.2  | 20.7  | 21.6  |
|                                                                        |             | MCDB   | 30.9  | 33.7  | 29.8  | 24.4  | 19.7  | 15.9  | 16.1  | 19.5  | 25.6  | 27.3  | 29.7  | 29.9  |
| (30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 21.4  | 21.6  | 19.7  | 17.2  | 14.9  | 13.0  | 12.2  | 12.6  | 14.2  | 16.7  | 19.2  | 20.0  |
|                                                                        |             | MCDB   | 31.0  | 31.1  | 28.7  | 24.6  | 19.8  | 15.5  | 15.2  | 18.4  | 23.3  | 26.5  | 27.7  | 28.4  |
|                                                                        | 5%          | WB     | 20.4  | 20.4  | 18.9  | 16.2  | 13.9  | 12.1  | 11.4  | 11.6  | 13.2  | 15.5  | 18.1  | 19.0  |
|                                                                        |             | MCDB   | 31.8  | 29.9  | 28.1  | 23.9  | 18.6  | 14.8  | 14.4  | 16.8  | 21.2  | 25.2  | 28.1  | 29.8  |
|                                                                        | 10%         | WB     | 19.3  | 19.2  | 17.9  | 15.2  | 13.2  | 11.3  | 10.6  | 10.8  | 12.2  | 14.4  | 16.8  | 17.9  |
|                                                                        |             | MCDB   | 32.0  | 29.8  | 28.1  | 23.1  | 17.5  | 14.0  | 13.5  | 15.5  | 19.3  | 24.0  | 27.3  | 30.0  |
| (36) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 15.7  | 15.0  | 13.5  | 12.3  | 9.9   | 8.6   | 8.7   | 10.5  | 12.8  | 14.7  | 15.1  | 15.4  |
|                                                                        |             | MCDBR  | 18.3  | 17.1  | 15.4  | 14.7  | 11.6  | 9.9   | 10.1  | 13.0  | 16.5  | 17.8  | 17.8  | 17.8  |
|                                                                        |             | MCWBR  | 5.7   | 5.4   | 5.1   | 5.8   | 5.4   | 5.4   | 5.6   | 6.2   | 6.8   | 6.7   | 5.9   | 5.8   |
|                                                                        | 5% WB       | MCDBR  | 14.6  | 13.3  | 13.2  | 11.8  | 9.6   | 8.5   | 8.8   | 11.2  | 14.0  | 15.0  | 14.4  | 13.5  |
|                                                                        |             | MCWBR  | 6.0   | 5.1   | 5.4   | 5.4   | 5.2   | 5.2   | 5.5   | 6.0   | 6.5   | 6.2   | 5.8   | 5.8   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.336 | 0.333 | 0.316 | 0.307 | 0.298 | 0.289 | 0.289 | 0.299 | 0.321 | 0.330 | 0.340 | 0.329 |
|                                                                        | taud        |        | 2.508 | 2.534 | 2.572 | 2.570 | 2.575 | 2.595 | 2.593 | 2.550 | 2.482 | 2.471 | 2.475 | 2.520 |
|                                                                        | Ebn at Noon |        | 1000  | 981   | 962   | 914   | 868   | 850   | 870   | 909   | 940   | 973   | 990   | 1011  |
|                                                                        | Edh at Noon |        | 113   | 106   | 94    | 84    | 73    | 67    | 71    | 84    | 101   | 111   | 116   | 112   |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 7.87  | 7.04  | 5.76  | 4.20  | 2.91  | 2.36  | 2.54  | 3.51  | 4.79  | 6.23  | 7.10  | 7.82  |
|                                                                        | RadStd      |        | 0.37  | 0.35  | 0.27  | 0.21  | 0.14  | 0.15  | 0.12  | 0.22  | 0.33  | 0.40  | 0.37  | 0.31  |

Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (12 neighbors) | +0.42 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +126    | +81     |

Nomenclature: See separate page